REACTING MASS CALCULATIONS 3

1) Calculate the mass of calcium that can react with 40 g of oxygen.

$2Ca + O_2 \rightarrow 2CaO$

moles O_2 = $\frac{mass}{M_r}$ = $\frac{40}{32}$ = 1.25 mol

moles Ca = 2 x 1.25 = 2.50 mol

mass Ca = M_r x moles = 40 x 2.50 = 100 g

2) Calculate the mass of fluorine that reacts with 3.9 g of potassium.

$2K + F_2 \rightarrow 2KF$

moles K = $\frac{mass}{M_r}$ = $\frac{3.9}{39}$ = 0.10 mol

moles F_2 = $\frac{0.10}{2}$ = 0.050 mol

mass F_2 = M_r x moles = 38 x 0.050 = 1.9 g

3) Calculate the mass of nitrogen that reacts with 30 g of hydrogen.

$N_2 + 3H_2 \rightarrow 2NH_3$

moles H_2 = $\frac{mass}{M_r}$ = $\frac{30}{2}$ = 15 mol

moles N_2 = $\frac{15}{3}$ = 5 mol

mass N_2 = M_r x moles = 28 x 5 = 140 g

4) What mass of carbon dioxide is made when 7.2 g of pentane (C_5H_{12}) burns in oxygen?

$C_5H_{12} + 8O_2 \rightarrow 5CO_2 + 6H_2O$

moles C_5H_{12} = $\frac{mass}{M_r}$ = $\frac{7.2}{72}$ = 0.10 mol

moles CO_2 = 5 x 0.10 = 0.50 mol

mass CO_2 = M_r x moles = 44 x 0.5 = 22 g

5) What mass of potassium can react with 4.0 g of oxygen?

$4K + O_2 \rightarrow 2K_2O$

moles O_2 = $\frac{mass}{M_r}$ = $\frac{4.0}{32}$ = 0.125 mol

moles K = 4 x 0.125 = 0.50 mol

mass K = M_r x moles = 39 x 0.5 = 19.5 g

6) What mass of chlorine reacts with 8.1 g of aluminium? $2Al + 3Cl_2 \rightarrow 2AlCl_3$

moles Al = $\frac{\text{mass}}{M_r}$ = $\frac{8.1}{27}$ = 0.30 mol

moles Cl_2 = $\frac{3}{2}$ x 0.30 = 0.45 mol

mass Cl_2 = $M_r\ x$ moles = 71 x 0.45 = 31.95 g

7) What mass of iron can be made from 20 kg of iron(III) oxide? $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

moles Fe_2O_3 = $\frac{\text{mass}}{M_r}$ = $\frac{20000}{160}$ = 125 mol

moles Fe = 2 x 125 = 250 mol

mass Fe = $M_r\ x$ moles = 56 x 250 = 14000 g

8) What mass of hydrogen is needed to react with 31.8 mg of copper(II) oxide? $CuO + H_2 \rightarrow Cu + H_2O$

moles CuO = $\frac{\text{mass}}{M_r}$ = $\frac{0.0318}{79.5}$ = 0.00040 mol

moles H_2 = 0.00040 mol

mass H_2 = $M_r\ x$ moles = 2 x 0.00040 = 0.00080 g

9) 5.95 g of hydrated cobalt(II) chloride decompose to form 3.25 g of anhydrous cobalt(II) chloride on heating. Calculate the formula mass of hydrated cobalt(II) chloride and the value of *n*. $CoCl_2.nH_2O \rightarrow CoCl_2 + nH_2O$

moles $CoCl_2$ = $\frac{\text{mass}}{M_r}$ = $\frac{3.25}{130}$ = 0.025 mol

mass H_2O = 5.95 – 3.25 = 2.70 g

moles H_2O = $\frac{\text{mass}}{M_r}$ = $\frac{2.70}{18}$ = 0.15 mol

Ratio of moles $CoCl_2 : H_2O$ = 0.025 : 0.15 = $\frac{0.025}{0.025}$ $\frac{0.150}{0.025}$ = 1 : 6

∴ *n* = 6 (nearest whole number)

Area	Strength	To develop	Area	Strength	To develop	Area	Strength	To develop
Done with care and thoroughness			Can find moles from mass			Can convert units		
Shows suitable working			Can use reacting ratios in equations			Can find water of crystallisation		
Can work out M_r			Can find mass from moles			Gives units		